

JKBOSE

2026 CLASS 11TH

BASIC MATH AND CALCULUS

(Revision)

→ 15 min

LECTURE-9

PHYSICS

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Topics To Be Covered

1 Revision

One
Short
15 min  

Copy saat main

phella 8 Lecture dekhe hona
chahiya

Full Focus.

Sab yaad hojaye gaa.

Recap From 1st Lec.

Maths

- Maths Bohat Aasaan hai
- number
- + - x ÷
- Fraction
- Angle / Power
- Trigonometry
- Graph / Equation
- $\frac{dy}{dx}$ $\int y dx$

Fraction

- $\frac{1}{2} = 0.5$
- $\frac{1}{3} = 0.33$
- $\frac{1}{4} = 0.25$
- $\frac{1}{10} = 0.1$
- $\frac{1}{100} = 0.01$
- $5.42 \times 100 = 542$

Power

- $2^2 = 2 \times 2$
- $x^0 = 1$
- $x^a \cdot x^b = x^{a+b}$
- $(x^a)^b = x^{a \times b}$

$$\frac{1}{0} = \infty$$

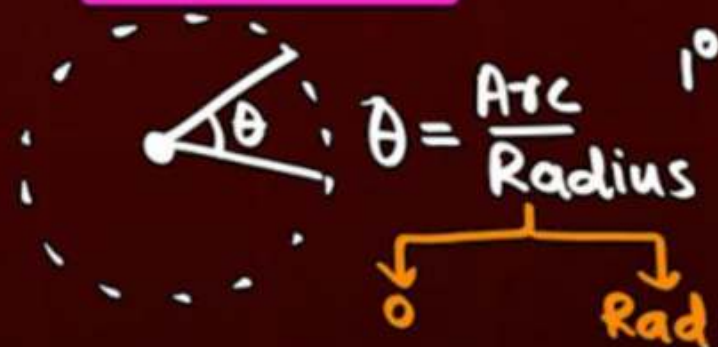
$$\frac{1}{\infty} = 0$$

$$\sqrt{2} = 2^{\frac{1}{2}}$$

$$\sqrt{2} = 1.4$$

$$\sqrt[3]{3} = 1.7$$

Angle θ



$$\rightsquigarrow 180^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$

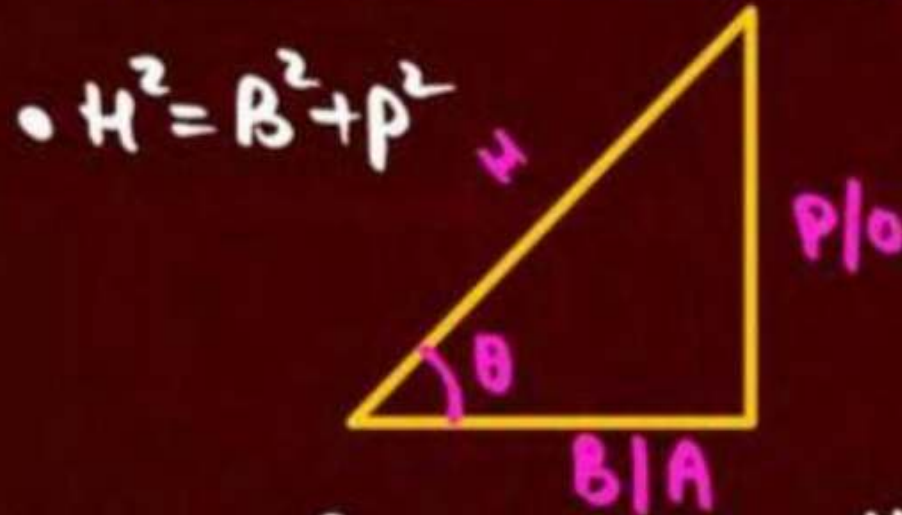
Recap From 2nd Lec.

Trigonometry Table

	0°	30°	45°	60°	90°	120°	135°	150°	180°
Angle	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

Trigonometry

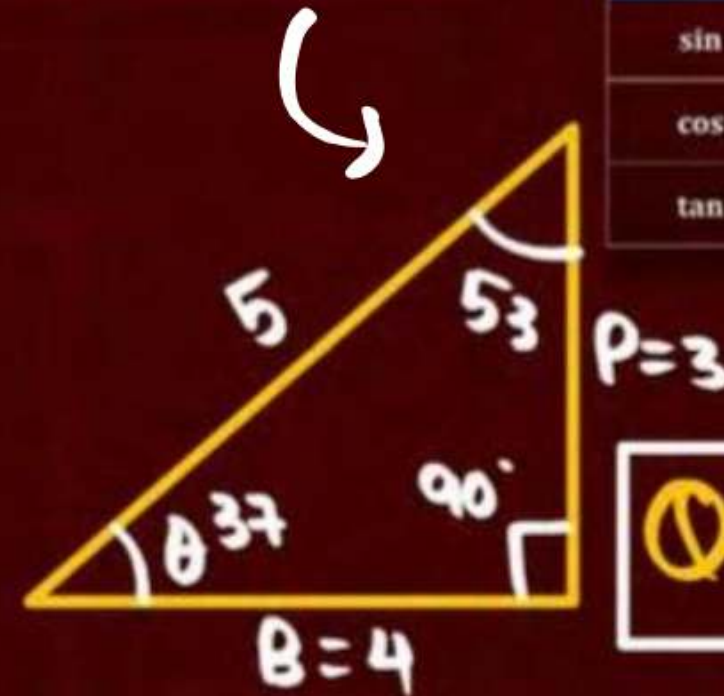
Special Δ



$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$



Quadrant System

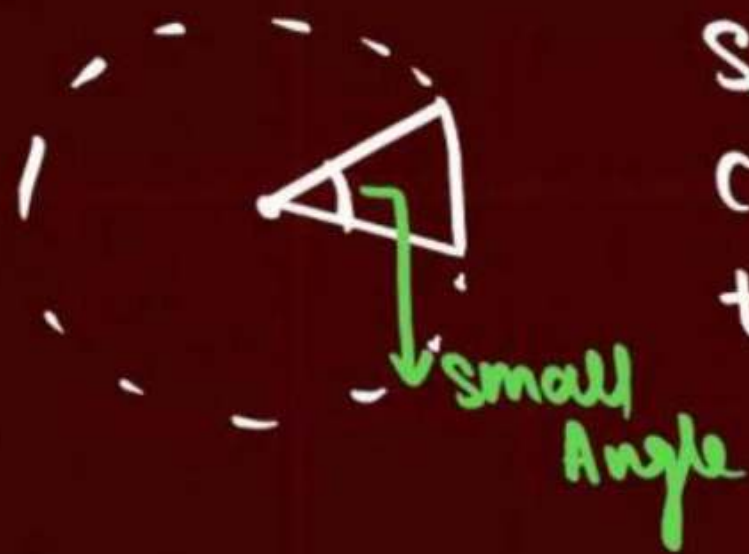


$$\sin^2 \theta + \cos^2 \theta = 1$$

Recap From 3rd Lec.



Small Angle Approx.



$$\begin{aligned}\sin \theta &= \theta \\ \cos \theta &= 1 \\ \tan \theta &= \theta\end{aligned}$$

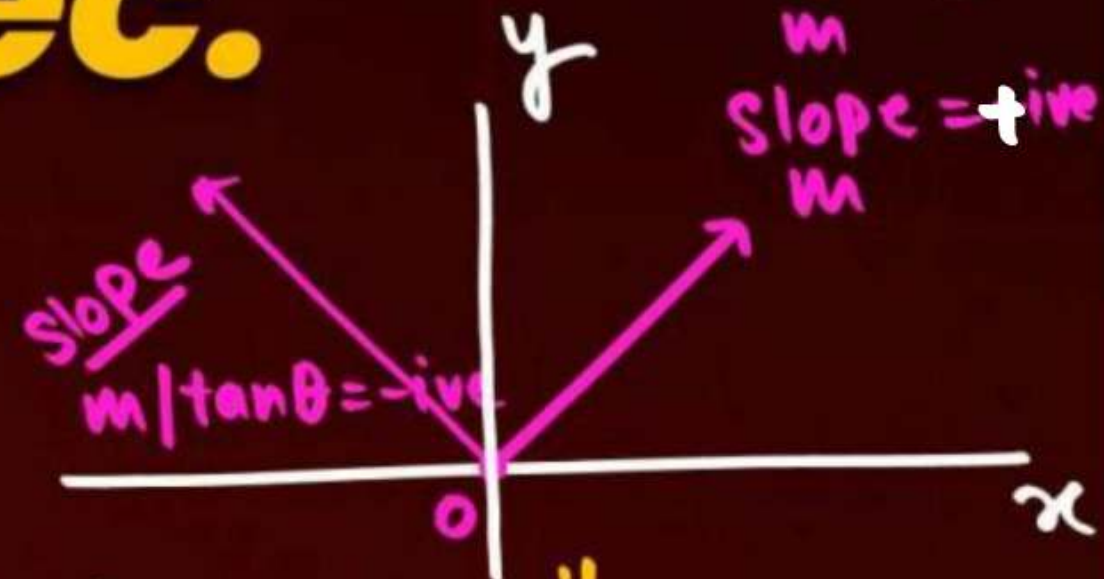
Equation

English Statement

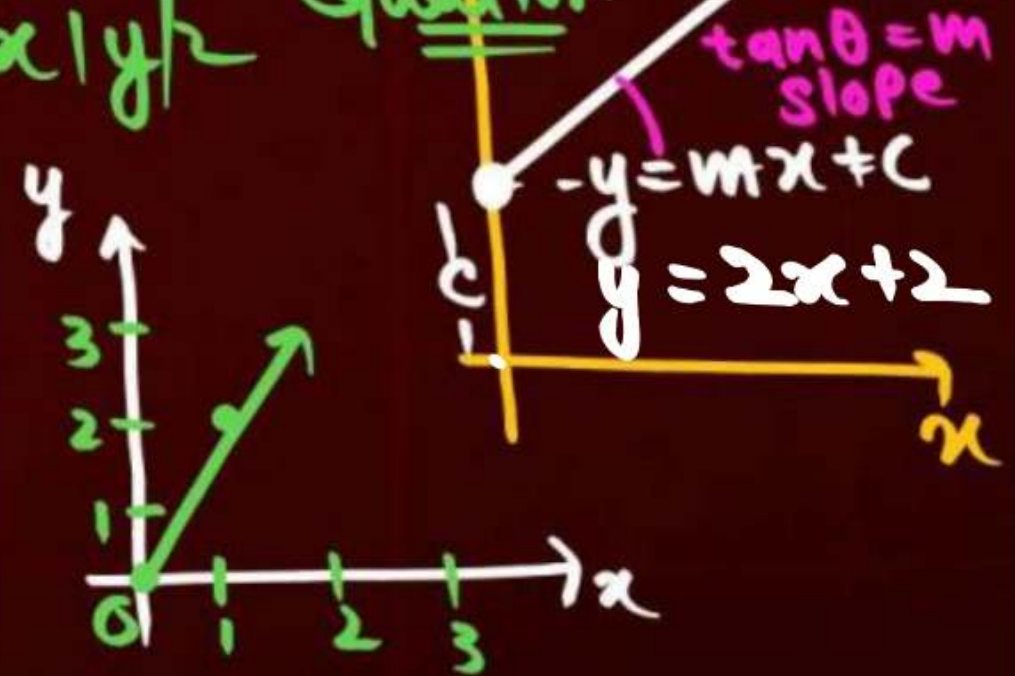
Slope $\frac{y}{x}$

$$y = \theta x$$

x	0	1	2
y	0	2	4



(i) Straight line equation



Recap From 4th Lec. 3rd



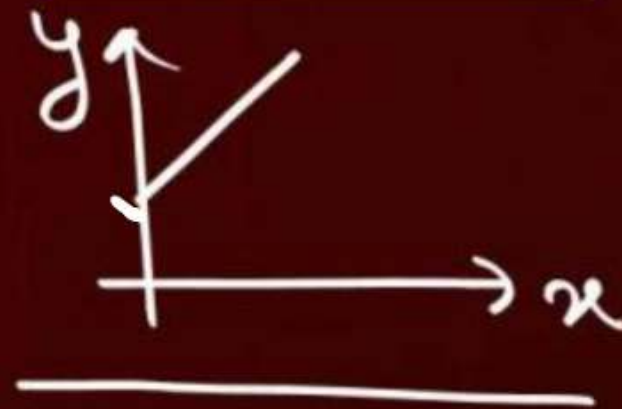
Equation

$$\underline{y = mx + c}$$

$$y = x^2$$

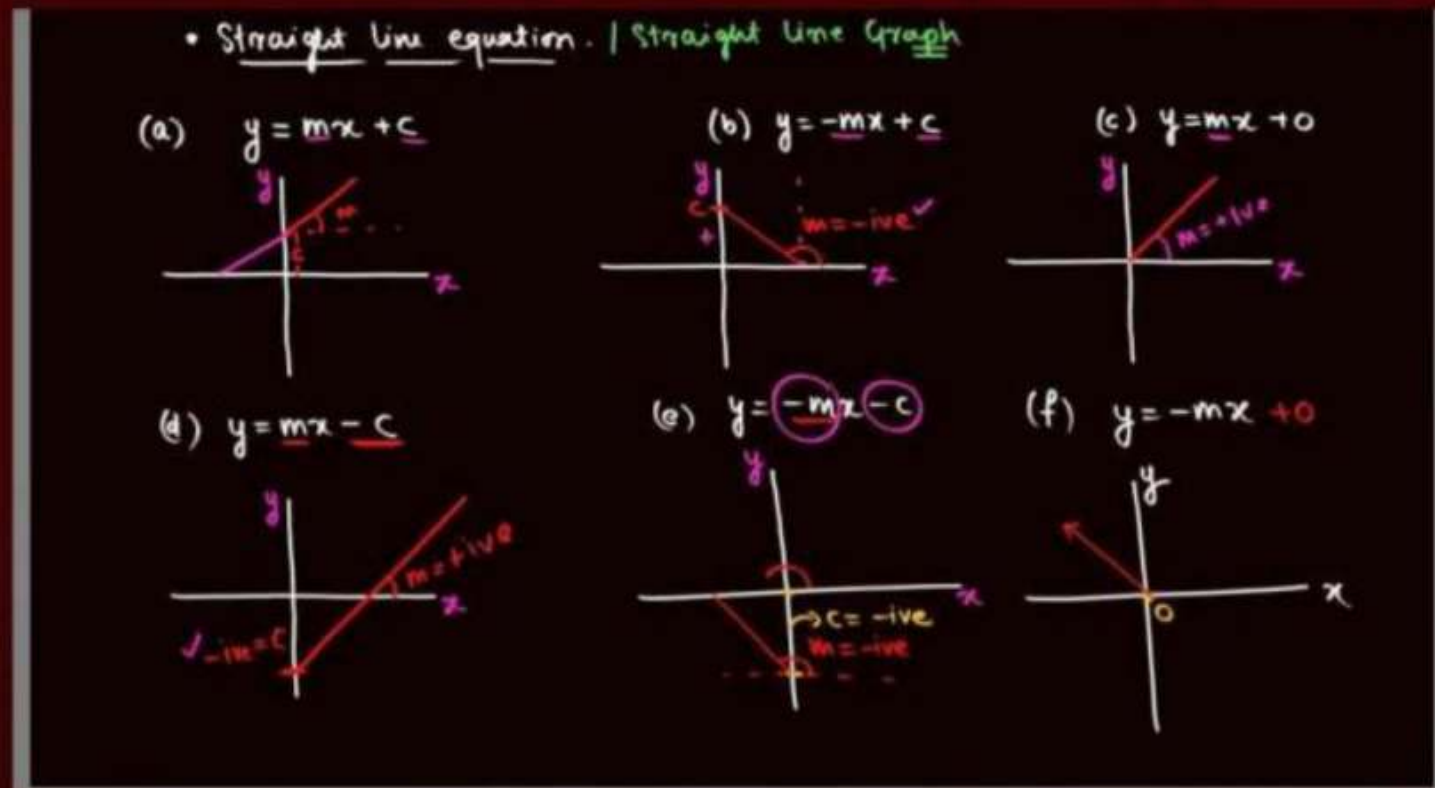
→ Table →

Graph

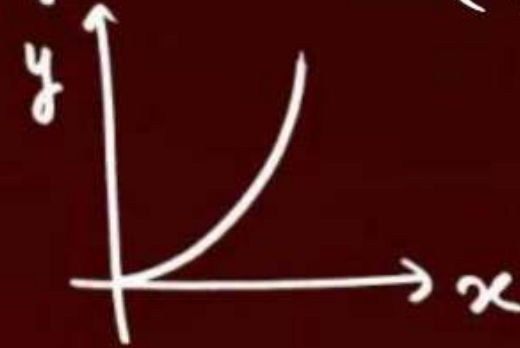


Recap From 5th Lec.

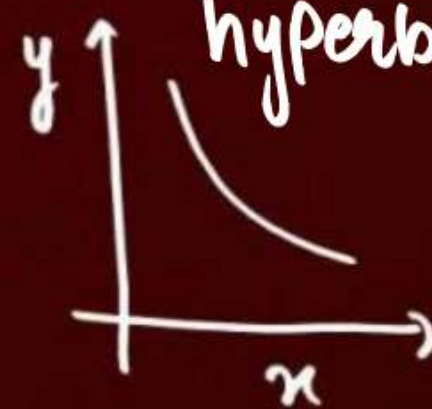
① $y = \underline{mx} + c$ (Important)



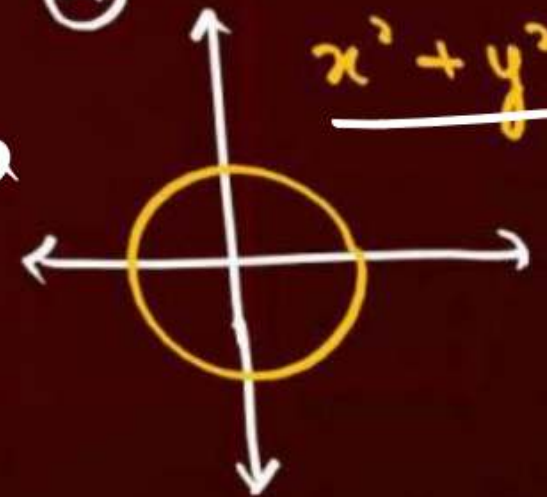
② $y = x^2$ (Parabolic graph)



③ $y = \frac{1}{x}$ hyperbola



④ $x^2 + y^2 = R^2$



Recap From 6th Lec.

Differentiation

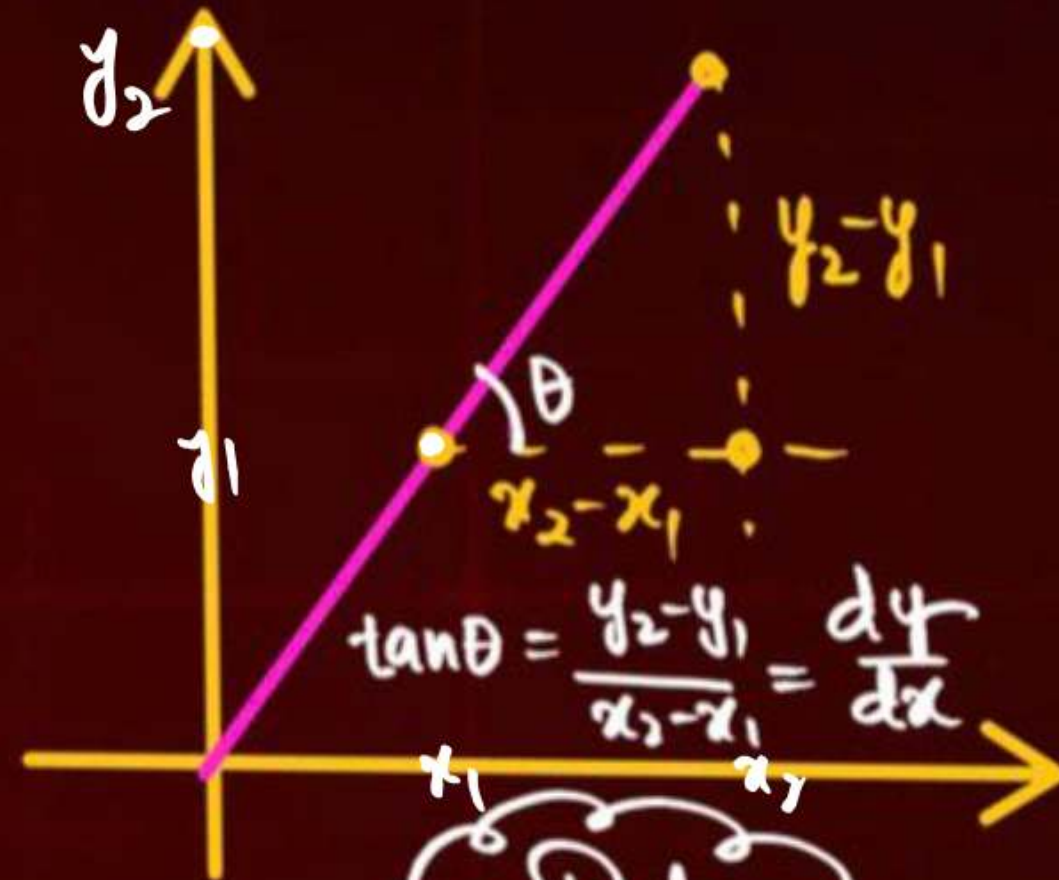
$$y = 2x$$

	x_1	x_2		
x	1	2	3	4
y	2	4	6	8
	y_1	y_2		

$$y = 2x$$

↓
Slope

differentiation $\left(\frac{dy}{dx}\right)$



$$\text{differentiation} = \frac{\text{Small change in } y}{\text{Small change in } x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{dy}{dx} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 2}{2 - 1} = \frac{2}{1} = 2$$

Rules

Recap from lecture. 7.



Rules for differentiation

$$\textcircled{1} \checkmark \frac{dc}{dx} = 0$$

$$\textcircled{2} \checkmark \frac{dx^n}{dx} = nx^{n-1}$$

$$\textcircled{3} \checkmark \frac{de^x}{dx} = e^x$$

$$\textcircled{4} \checkmark \frac{d \ln x}{dx} = \frac{1}{x}$$

$$\textcircled{5} \checkmark \frac{d \sin x}{dx} = \cos x$$

$$\textcircled{6} \checkmark \frac{d \cos x}{dx} = -\sin x$$

$$\textcircled{7} \checkmark \frac{d \tan x}{dx} = \sec^2 x$$

$$\textcircled{8} \checkmark \frac{d \csc x}{dx} = -\csc x \cot x$$

$$\textcircled{9} \checkmark \frac{d \sec x}{dx} = \sec x \tan x$$

$$\textcircled{10} \checkmark \frac{d \cot x}{dx} = -\operatorname{cosec}^2 x$$

$$\textcircled{11} \checkmark \frac{d(u+v+w)}{dx} = \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx}$$

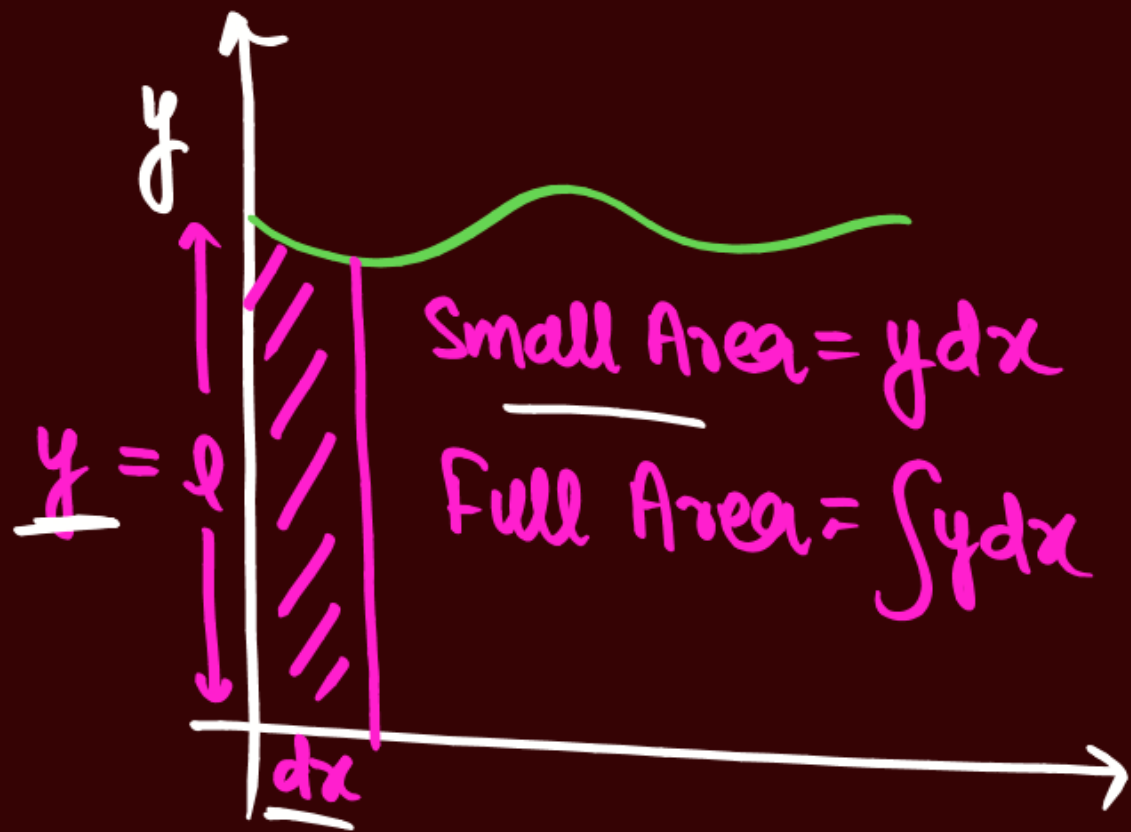
$$\textcircled{12} \checkmark \frac{d(u \cdot v)}{dx} = v \frac{dv}{dx} + u \frac{du}{dx}$$

$$\textcircled{13} \checkmark \frac{d \frac{u}{v}}{dx} = \frac{v \frac{dv}{dx} - u \frac{du}{dx}}{v^2}$$

Recap lecture no. 8.

Integration

- Formula
- Area under graph



Rule - ①

$$\textcircled{1} \int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\textcircled{2} \int \sin x dx = -\cos x$$

$$\textcircled{3} \int (u+v+w) dx = \int u dx + \int v dx + \int w dx$$

$$\textcircled{4} \int (u-v-w) dx = \int u dx - \int v dx - \int w dx$$

$$\textcircled{5} \int (u+v+w) dx = \int u dx + \int v dx - \int w dx$$

Chain Rule $\left(\frac{dy}{dx}\right) = \frac{d \sin x^2}{dx} = 2x^{2-1} \cos x^2 = 2x^1 \cos x^2$

Integration

$$\int y dx \quad \text{from } x_1 \text{ to } x_2 \quad \int_{x_1}^{x_2} y dx$$

Thank
You